Parabolic Flight, Inertia and the Representations of Gravitation

Free fall, counter-path, light and horizon in the time-mass duality

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Parabolic Flight, Inertia and the Representations of Gravitation

Abstract

In parabolic flight, gravity disappears for the passengers. This document starts from that question and reads it with the time-mass duality $\tilde{T} \cdot m = 1$: free fall is the path of stationary phase of a mode with position-dependent mass, and weight on the ground is the deviation of the rest world line from that path. What cannot be transformed away are the tides. Gravitation appears in three representations: as space curvature, as a change of mass or clock rate, and as fractal path lengthening. They are mathematical models, not statements about whether space is “really” flat or curved, and they are equivalent for the static, non-rotating weak field and for the conformally flat part of the geometry; how far the equivalence reaches beyond that is examined in Doc. 392. For light, clock rate and path are complementary halves: each alone gives $0.875''$ at the solar limb, both together $1.75''$ and $\gamma = 1$. The purely conformally flat metric gives no deflection. With the logarithmic form of the mass change, $\beta = 1$ follows as well, and with it the perihelion shift of GR. At the horizon, T_0 limits the comparison with a distant reference, not the free crossing. A separate quantisation of gravitation is not needed.

Status markers

[B] Proved algebraically.
[K] Confirmed numerically (verification script, comparison with measured values).
[S] Setting, open or speculative.
[X] Refuted or not viable.
[Q] Convention.

Measured values: Pound–Rebka 1960, Cassini (Bertotti et al. 2003), IAU and CODATA 2022 constants. Verification script: `pruef_389_parabelflug.py`.

1 The question

In parabolic flight the aircraft flies a ballistic parabola for about twenty seconds. The passengers float, although the Earth attracts them as before. On the ground, by contrast, they feel their weight, although they are not moving. The question is what this weight is, if it disappears in free fall.

The answer this document works out: the force we feel as weight exists only because we deviate from the path we would take without support. The ground forces us onto a path that is accelerated exactly opposite to the path of fall. In parabolic flight this deviation drops out, and with it the weight. The following sections recalculate this in the representation of FFGFT and place it among the different readings of gravitation.

2 Free fall in the mass representation

In the representation “space flat, mass variable” (Doc. 078) a mode in the potential Φ carries the position-dependent mass $m(x)$. Its action is that of a clock ticking at the rate $m(x)c^2$:

$$L = -m(x)\,c^2\sqrt{1 - v^2/c^2}, \qquad m(x) = m_0\, e^{\Phi(x)/c^2}. \tag{1}$$

Here $m(x)$ is the mass as compared from the distant reference. A clock at Φ ticks in that reference’s time at

$$\omega(\Phi) = \omega_0\sqrt{g_{00}} \approx \omega_0\Big(1 + \frac{\Phi}{c^2}\Big), \qquad m(x) = m_0\sqrt{g_{00}}, \qquad g_{00} = \Big(\frac{m}{m_0}\Big)^2 = e^{2\Phi/c^2} \quad \textbf{[B]}, \tag{2}$$

since with $\tilde{T}\cdot m = 1$ the mass is the frequency of the mode. $m(x)$ is the Killing energy of the mode; measured with a co-moving clock the mass remains m_0. For small velocity the Euler–Lagrange equation gives

$$\ddot{\vec{x}} = -c^2\,\nabla \ln m = c^2\,\nabla \ln \lambda_4 = -\nabla\Phi \quad \textbf{[B]}, \tag{3}$$

with the mode length $\lambda_4 = 2\pi/m$ (Doc. 314). Without a metric and without Christoffel symbols, the fall follows from the gradient of the mass alone: bodies fall towards where their mass is smaller and their mode length larger. Since $\nabla \ln m$ is the same for every mode, all bodies fall alike; in this form the equivalence principle is built in.

At the Earth’s surface $g/c^2 = 1.09\times10^{-16}$ per metre **[K]**. This is the same number by which a clock runs faster per metre of height. In this representation, gravitational acceleration and clock rate are a single quantity.

Sign. The direction of the mass change is fixed by the redshift. Light rising in a gravitational field arrives redshifted; the Pound–Rebka experiment measured this over $22.5\,\mathrm{m}$ as $(2.57 \pm 0.26) \times 10^{-15}$, with 2.46×10^{-15} expected **[K]**. In the mass representation this means: the mode below, compared from the upper reference, has the smaller mass; m grows with Φ. With this sign the body falls downwards. The sign is therefore not an open choice but determined by measurement.

3 Rest as a counter-path

The phase of a mode along a path is $\varphi = -\int m(x)\, c^2\, d\tau$. Free fall is the path on which this phase is stationary, $\delta\varphi = 0$. The rest world line $\vec{x} = \text{const}$ is not: its variation does not vanish but leaves the remainder

$$\left.\frac{\partial L}{\partial \vec{x}}\right|_{v=0} = -m\,\nabla\Phi = -m\,\vec{g} \quad \textbf{[B]}. \tag{4}$$

For a body to be at rest nonetheless, the ground must balance exactly this remainder with $+m\vec{g}$ upwards. That is the weight. It is not an additional force on top of gravity, but the measure of how far the rest world line deviates from the path of stationary phase. In the language of the mode it is a phase detuning: the mode at rest runs against the rate it would have in free fall.

In parabolic flight the aircraft follows the path of fall. The deviation vanishes, and with it the weight. What the passengers experience as heaviness on the ground and as weightlessness in the aircraft is the same path, once with and once without the counter-path. As a formula:

$$\vec{a}_{\text{fall}} = c^2\,\nabla \ln \lambda_4, \qquad \vec{a}_{\text{inert}} = -c^2\,\nabla \ln \lambda_4 = -\vec{a}_{\text{fall}} \quad \textbf{[B]}. \tag{5}$$

On the ground the two cancel; in parabolic flight the counter-path is zero. In this sense the inertial force exists only because we have an exactly opposite path.

4 What cannot be transformed away

The gradient $\nabla \ln m$ can be transformed away locally: in the freely falling frame it is zero, as are the Christoffel symbols in the local inertial frame of GR. The second derivative cannot be transformed away. The tides are the Hessian

$$\partial_i\partial_j\Phi = c^2\,\partial_i\partial_j \ln m, \qquad \text{eigenvalues } (-2,\,1,\,1)\,\frac{GM}{r^3} \text{ on the axis}, \qquad \nabla^2 \ln m = \frac{4\pi G\rho}{c^2} \quad \textbf{[B]}. \tag{6}$$

In vacuum the trace is zero; inside a mass it gives the Poisson equation, now as an equation for the mass field. Outside, $m = m_0\, e^{-GM/(rc^2)} \approx m_0\big(1 - GM/(rc^2)\big)$; the gradient $\partial_r \ln m = GM/(r^2c^2)$ can be transformed away, the second derivative $c^2\,\partial_r^2 \ln m = -2GM/r^3$ cannot **[B]**. At the Earth's surface $2GM/r^3 = 3.08 \times 10^{-6}\,\mathrm{s}^{-2}$ **[K]**. This remainder persists in parabolic flight as well: two bodies at different places in the aircraft do not fall exactly alike. What the mass representation calls "curvature" is the curvature of the mass field, not that of a spatial geometry.

5 Three representations

A position-dependent mass is equivalent to a position-dependent time, since $\tilde{T} \cdot m = 1$. What one calls it is a choice of representation:

- **Space curvature:** mass constant, space and time curved, as in GR.

- **Change of mass or clock rate:** space flat, mass and clock rate position-dependent (Doc. 078).
- **Fractal path lengthening:** space flat, propagation paths geometrically stretched (Docs. 026, 041, 182).

These are mathematical models that must agree by virtue of the duality insofar as they yield the same observables. Ontologically, one can say neither that space is flat nor that it is curved; mixed forms are conceivable. All that matters is that the observables agree. This has been shown for the static, non-rotating weak field with $\gamma = \beta = 1$ and for the conformally flat part, to which the universe at large also belongs. Around rotating masses and for gravitational waves the flat representation must additionally supply a co-moving vector field and a tensor part; in the curvature representation both are contained through the field equations (Docs. 312, 392).

Light as the test. For clocks, redshift and slow bodies the clock rate suffices. Light tests more, because it also feels the spatial geometry. In Fermat form light propagates through a refractive index $n = 1 + (1+\gamma)\,GM/(rc^2)$; the deflection at the solar limb and the Shapiro delay scale with $(1+\gamma)/2$:

Representation	γ	Deflection at the solar limb
purely conformally flat metric (Doc. 143)	-1	0
clock rate only (Einstein 1911)	0	$0.875''$
clock rate and fractal path lengthening (Doc. 308)	1	$1.751''$

[K]. Cassini measures $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$; only the last row is compatible with the measurement. Doc. 308 computes the same by ray tracing: clock rate and path lengthening give $0.875''$ each, together $1.750''$. They are complementary halves, not alternatives.

This sharpens the three representations. “Space flat, mass variable” on its own carries only the temporal half. The flat representation becomes complete only with the fractal path lengthening, which carries the spatial half. The space curvature of GR is then equivalent to clock rate and path together. The temporal half is measured by clocks and is the same in every representation; the spatial half can be written as curvature, as path lengthening, or as a mixture of both. The freedom of representation therefore lies in the spatial half, and the condition for every mixed form is $\gamma = 1$.

The conformally flat metric $g_{\mu\nu} = \eta_{\mu\nu}(1 - 2\xi E_{\text{field}}/E_P)$ of Doc. 143 is none of these representations: it stretches space and time by the same factor, null geodesics remain unchanged, and light propagates as in flat space. This is the Nordström case (Doc. 143). The same holds for the conformal coupling $g_{\mu\nu} \to \Omega^2 g_{\mu\nu}$ of the matter fields to the time field (A142): the Maxwell field is conformally invariant in four dimensions, and the coupling leaves light unaffected **[B]**. For massive matter it supplies the clock rate, i.e. the temporal half; the spatial half has to come from the path lengthening (A142).

6 Second order: the perihelion shift

Light deflection tests the first order in the potential. The perihelion shift additionally tests the second order of the time component, the parameter β in $g_{00} = 1 - 2U + 2\beta U^2$ with $U = GM/(rc^2)$. The factor relative to GR is $(2 + 2\gamma - \beta)/3$. In the mass representation $g_{00} = (m/m_0)^2$, and β depends on the form of $m(\Phi)$:

Form	β	with γ	Mercury (″/cent.)
$m = m_0\, e^{\Phi/c^2}$	1	1	42.98 (GR)
$m = m_0\,(1+\Phi/c^2)$	$\frac{1}{2}$	1	50.1
Nordström	$\frac{1}{2}$	-1	-7.2
clock rate only	1	0	14.3

[K]. The logarithmic form $\ln(m/m_0) = \Phi/c^2$ is the same one that in Section 2 gives the fall $\ddot{\vec{x}} = -\nabla\Phi$ exactly and not merely approximately. With it, and with the spatial half from Doc. 308, the flat representation gives the perihelion shift of GR. The logarithmic form itself is set, not derived from $\tilde{T}\cdot m = 1$ **[S]**; the linear form is excluded by the Mercury measurement.

7 The relation $\lambda_4 \cdot m = 2\pi$ holds locally

The position relation $\lambda_4 \cdot m = 2\pi$ (Doc. 314) connects proper quantities: the mode length as a co-moving rod measures it, and the mass as a co-moving clock measures it. In coordinates of the distant reference it does not hold unchanged once the spatial half is written as curvature, because coordinate lengths then carry a factor of the spatial metric. In the isotropic form of GR,

$$g_{ij} = \Big(1 - \frac{2\Phi}{c^2}\Big)\delta_{ij}, \qquad \ell_{\text{coord}} = \ell\Big(1 + \frac{\Phi}{c^2}\Big), \qquad \lambda_{4,\text{coord}} = \frac{2\pi}{m(x)} = \lambda_4\Big(1 - \frac{\Phi}{c^2}\Big) \quad \textbf{[B]}. \tag{7}$$

Seen from the distant reference, spatial lengths deep in the potential appear shortened and the temporal mode length lengthened: the two factors run against each other. In the purely conformally flat metric both would scale alike with $1/m$, and light would remain undeflected. The spatial factor is exactly the half that light additionally feels. In the representation with fractal path lengthening it sits in the stretched path, as a refractive contribution $n - 1 = -\Phi/c^2$ added to the clock rate. The gradient $\nabla \ln \lambda_4$ in Section 2 is accordingly a local quantity: in the freely falling frame it vanishes, in the frame at rest it is g/c^2.

8 Horizon: comparison and crossing

Doc. 306 describes the horizon as an extremum: there T reaches the value $T_0 = \xi t_P$ and the energy per excitation the value E_P/ξ. It adds that the "slowest time" at the horizon is a statement about the comparison, not about experience. The calculation confirms this.

A static observer at r measures, for a mode of energy E (referred to the distant reference), the local energy $E/\sqrt{1 - r_s/r}$; it diverges at the horizon **[B]**. The bound $E \le E_P/\xi$ cuts this approach off at the proper distance $\rho_{\min} = 2\, r_s\, \xi\, E/E_P$ **[B]**. A freely falling observer, by contrast, measures $E/(1 + \sqrt{r_s/r})$ for incoming light, hence $E/2$ at the horizon, which is finite **[B]**. For this observer there is nothing to bound at the horizon.

The bound T_0 therefore concerns the static reading, the comparison with a distant reference. The free crossing is not affected; this is the same distinction as between rest and fall in parabolic flight. Whether T_0 additionally bounds the radius in the interior from below requires a dynamics of the interior that the corpus does not supply. The physical validity in the black-hole regime remains open **[S]** (Doc. 384).

9 Two masses, two readings of G

Two masses at the upper end are easily confused:

- $M_{\text{coll}} = m_P/\sqrt{2}$: here the Schwarzschild radius equals the reduced Compton length **[B]** (Docs. 325, 329).
- $m_P/\xi = 7500\, m_P \approx 1.63 \times 10^{-4}$ kg: the dual upper bound per excitation, belonging to $L_0 = \xi \ell_P$ **[K]** (Docs. 290, 306).

Both are correct but different; their ratio is $7500\sqrt{2} \approx 10\,607$. According to Doc. 306, both bounds hold per excitation, not per aggregate: a star is not a single winding, and $\hbar/(Mc^2)$ for a stellar mass is not a state of the relation.

The same applies to G. With the measured, fixed G, $m \cdot G = \xi^2/4$ holds for the characteristic mass m_{char} of the anchor (R58, A142). For arbitrary m the relation holds only if G runs along, as Doc. 350 reads it for an evaporating hole. The statement "no mass singularity with finite G" is, in the first reading, a consequence of the definition; in the second, a statement about the dynamics.

10 No separate quantisation of gravitation

In FFGFT gravitation is not an additional field and needs no graviton. It is already contained in the time-field geometry and shows itself as an effect in the redshift, read as space curvature, as mass change or as fractal path lengthening. A separate quantisation of gravitation is therefore not required (R58, A142). What remains open is the transfer of the classical equations into loop corrections for situations beyond the Schwarzschild limit (A142).

11 What remains open

- the logarithmic form $\ln(m/m_0) = \Phi/c^2$, which gives $\beta = 1$, as a consequence of the duality rather than only as a setting **[S]**;
- the origin of the fractal path lengthening as the spatial half from the structure of the carrier; Doc. 308 gives the K3 area invariance as the reason, confirmed numerically but not proved **[K]**;
- the dynamics inside a horizon and the validity of $m \cdot G = \xi^2/4$ beyond the anchor **[S]**.

12 Conclusion

In parabolic flight it is not gravitation that disappears but the counter-path. Free fall is the path of stationary phase of a mode with position-dependent mass, weight on the ground is the deviation of the rest world line from that path, and only the tides persist in every frame. Space curvature, mass change and fractal path lengthening are models of the same situation, not statements about the nature of space; they are equivalent for the static, non-rotating weak field and the conformally flat part, while dragging and gravitational waves require additional parts in the flat representation (Doc. 392). For light, clock rate and path are complementary halves; only both together give $1.75''$ and $\gamma = 1$. With the logarithmic form of the mass change the perihelion shift of GR follows as well. At the horizon, the same distinction between rest and fall separates the comparison with a distant reference from the free crossing.

Verification script: `2/python/Dok389_Skripte/pruef_389_parabelflug.py` — 34/34.